Advanced Causal Inference

Lecture 06: IPW, Balancing Estimators

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This lecture note is partially based on the lecture notes/textbook of Hyunseung Kang, Ying Zhou, Hernán and Robins (2020)

Outline

1. Observational Studies
2. Preliminary
3. Propensity Score-Based Estimators
4. Balancing Estimators
5. Assess Overlap
6. Assignment

Causal Assumptions in Observational Studies

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

Assumption (Conditional Ignorability)

- $Y(a) \perp\!\!\!\perp A \,|\, X,\ \forall a$

Assumption (Positivity/Overlap)

- $\Pr(A = a \,|\, X) \in (0, 1),\ \forall (a, X)$

- Under these assumptions,

$$\begin{aligned}
\tau_{\mathsf{ATE}} &= \mathrm{E}\{Y(1) - Y(0)\} \\
&= \mathrm{E}\big\{\mathrm{E}(Y|A = 1, X) - \mathrm{E}(Y|A = 0, X)\big\} && \text{OR-based Representation} \\
&= \mathrm{E}\left\{\frac{AY}{\Pr(A = 1|X)} - \frac{(1 - A)Y}{\Pr(A = 0|X)}\right\} && \text{IPW-based Representation}
\end{aligned}$$

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M-Estimator

- van der Vaart (1998, Chapter 5) and Stefanski and Boos (2002)
- MLE, OLS, etc... are instances of the M-estimator
- $\Psi(O;\theta)$: moment function satisfying $\mathrm{E}\{\Psi(O;\theta^*)\} = 0$ with $\dim(\Psi) = \dim(\theta)$
- The M-estimator of $\widehat{\theta}$ solves $0 = \dfrac{1}{N}\sum_{i=1}^{N} \Psi(O_i;\widehat{\theta})$
- Under certain regularity conditions, we have

$$\sqrt{N}(\widehat{\theta} - \theta^*) \xrightarrow{D} N(0, \Sigma_1^{-1}\Sigma_2\Sigma_1^{-\intercal}) \ , \quad \begin{cases} \Sigma_1 = \mathrm{E}\{\nabla_\theta \Psi(O;\theta^*)\} \\ \Sigma_2 = \mathrm{E}\{\Psi(O;\theta^*)\Psi^\intercal(O;\theta^*)\} \end{cases}$$

- A consistent variance estimator is given by $\widehat{\Sigma}_1^{-1}\widehat{\Sigma}_2\widehat{\Sigma}_1^{-\intercal}$:

$$\widehat{\Sigma}_1 = \frac{1}{N}\sum_{i=1}^{N} \{\nabla_\theta \Psi(O_i;\widehat{\theta})\} \ , \quad \widehat{\Sigma}_2 = \frac{1}{N}\sum_{i=1}^{N} \{\Psi(O_i;\widehat{\theta})\Psi^\intercal(O_i;\widehat{\theta})\}$$

- We can construct a confidence interval based on the asymptotic normality

GMM-Estimator

- Hansen (1982): generalized method of moments (GMM) estimators
- Suppose $m = \dim(\Psi) \geqslant \dim(\theta) = p$ where $\mathrm{E}\{\Psi(O;\theta^*)\} = 0$
- The GMM estimator is given by

$$\widehat{\theta} = \arg\min_{\theta} \left\{ \frac{1}{N} \sum_{i=1}^{N} \Psi(O_i;\theta) \right\}^{\intercal} \Omega \left\{ \frac{1}{N} \sum_{i=1}^{N} \Psi(O_i;\theta) \right\} , \quad \Omega \in \mathbb{R}^{m \times m} : \text{s.p.d.}$$

- Under certain regularity conditions, we have

$$\sqrt{N}\big(\widehat{\theta} - \theta^*\big) \xrightarrow{D} N\Big(0, (\Sigma_1^{\intercal}\Omega\Sigma_1)^{-1}(\Sigma_1^{\intercal}\Omega\Sigma_2\Omega\Sigma_1)(\Sigma_1^{\intercal}\Omega\Sigma_1)^{-\intercal}\Big) , \quad \begin{cases} \Sigma_1 = \mathrm{E}\big\{\nabla_{\tau,\eta}\Psi(O;\tau^*,\eta^*)\big\} \in \mathbb{R}^{m \times p} \\ \Sigma_2 = \mathrm{E}\big\{\Psi(O;\tau^*,\eta^*)\Psi^{\intercal}(O;\tau^*,\eta^*)\big\} \in \mathbb{R}^{m \times m} \end{cases}$$

- If $m = p$, the asymptotic variance simplifies to that of the M-estimator
- A plug-in variance estimator is consistent

Nonparametric Bootstrap

- Nonparametric bootstrap

Algorithm 1: Bootstrap procedure

Input: Sample $\mathcal{D} = (O_1, \ldots, O_N)$ where $O_i = (Y_i, A_i, X_i)$, number of replicates B

Output: Bootstrap Estimates

for $b = 1, \ldots, B$ **do**

  Sample $\mathcal{D}^{[b]}$ with replacement from $\mathcal{D}$;

  Compute $\widehat{\theta}^{[b]}$ from $\mathcal{D}^{[b]}$;

end

return $\{\widehat{\theta}^{[1]}, \ldots, \widehat{\theta}^{[B]}\}$;

- A bootstrap CI can be obtained from the quantiles of $\{\widehat{\theta}^{[1]}, \ldots, \widehat{\theta}^{[B]}\}$

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Review: Propensity Score

- The **propensity score** is defined as $e(X) = \Pr(A = 1 \mid X)$
- First introduced by Rosenbaum and Rubin (1983): Biometrika's most cited paper ($>$ 40K citations)
- **Property 1**: The propensity score is the balancing score: $A \perp\!\!\!\perp X \,|\, e(X)$
- **Property 2**: The propensity score is the optimal balancing score,
  i.e., for any balancing score $b(X)$, there exists a function f s.t. $e(X) = f(b(X))$
- **Property 3**: $Y(a) \perp\!\!\!\perp A|X \Rightarrow Y(a) \perp\!\!\!\perp A|e(X)$
- **Property 4**: $\tau_{\mathsf{ATE}} = \mathrm{E}[\mathrm{E}\{Y|A = 1, e(X)\} - \mathrm{E}\{Y|A = 0, e(X)\}]$
- Stratification/Subclassification is based on **Property 4**

Stratification/Subclassification

- The stratification estimator
  - Stratify units into subclasses based on the estimated propensity score
  - Directly compare treated and control units within the same subclass
  - Average the treatment effect over subclasses

- Formally, let
  - Estimate the propensity score, $\widehat{e}(X) = \widehat{\Pr}(A = 1 \mid X)$, using regression, machine learning, etc
  - $[\widehat{e}_{\text{min}}, \widehat{e}_{\text{max}}] = \texttt{range}\{\widehat{e}_i = \widehat{e}(X_i) | i = 1, \ldots, N\}$ be the range of estimated propensity scores
  - $\widehat{C}_k = [\widehat{q}_{k-1}, \widehat{q}_k)$ $(k = 1, \ldots, K)$ be disjoint divisions of the interval $[\widehat{e}_{\text{min}}, \widehat{e}_{\text{max}}]$
  - $\mathcal{I}_k = \{i \,|\, \widehat{e}_i \in \widehat{C}_k\}; \quad N_k = |\mathcal{I}_k|; \quad N_{ak} = |\mathcal{I}_k \cap \mathbb{1}(A = a)|$

- Then the stratification estimator is

$$\widehat{\tau}_k = \sum_{i \in \mathcal{I}_k} \left(\frac{A_i}{N_{1k}} - \frac{1 - A_i}{N_{0k}} \right) Y_i \ , \quad \widehat{\tau}_S = \sum_{k=1}^{K} \frac{N_k}{N} \widehat{\tau}_k$$

Stratification/Subclassification

- Note that

$$\mathrm{E}[\mathrm{E}\{Y \mid e(X), A=a\}] \simeq \sum_{k=1}^{K} \Pr\{e(X)=e_k\}\mathrm{E}\{Y \mid e(X)=e_k, A=a\} \simeq \sum_{k=1}^{K} \frac{N_k}{N}\frac{1}{N_{ak}} \sum_{i\in\mathcal{I}_k} \mathbb{1}(A_i=a)Y_i$$

- Therefore, $\widehat{\tau}_S$ is a histogram approximation to $\tau_{\mathsf{ATE}} = \mathrm{E}[E\{Y \mid e(X), A=1\} - \mathrm{E}\{Y \mid e(X), A=0\}]$

$$\widehat{\tau}_S = \sum_{k=1}^{K} \frac{N_k}{N} \sum_{i\in\mathcal{I}_k} \left(\frac{A_i}{N_{1k}} - \frac{1-A_i}{N_{0k}}\right) Y_i \simeq \mathrm{E}[\mathrm{E}\{Y \mid e(X), A=1\} - \mathrm{E}\{Y \mid e(X), A=0\}] = \tau_{\mathsf{ATE}}$$

- Another interpretation:

$$\begin{aligned}
\widehat{\tau}_S &= \sum_{k=1}^{K} \frac{1}{N} \sum_{i\in\mathcal{I}_k} \left(\frac{A_i}{N_{1k}/N_k} - \frac{1-A_i}{N_{0k}/N_k}\right) Y_i \\
&\simeq \frac{1}{N} \sum_{k=1}^{K} \sum_{i\in\mathcal{I}_k} \left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{1-A_i}{1-\widehat{e}(X_i)}\right\} Y_i \\
&= \frac{1}{N} \sum_{i=1}^{N} \left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{1-A_i}{1-\widehat{e}(X_i)}\right\} Y_i \\
&\simeq \mathrm{E}\left[\left\{\frac{A_i}{e(X_i)} - \frac{1-A_i}{1-e(X_i)}\right\} Y_i\right] = \tau_{\mathsf{ATE}}
\end{aligned}$$

How many subclasses?

- If K is too small, then cond. ign. does not hold conditional on $e(X)$ even approximately
- If K is too large, then we do not have enough units within each stratum of the estimated propensity score and many strata have only treated or control units
- Rosenbaum and Rubin (1983) recommends $K = 5$ and stratify on sample quantiles
- Leads to residual bias
- It is thus reasonable to increase K for large datasets as long as each stratum still has enough treated and control units
- The optimal choice of K and endpoints remain an open problem

Variance estimation

- Do as in a stratified randomized experiment:

$$\widehat{\mathrm{Var}}(\widehat{\tau}_S) = \sum_{k=1}^{K} \left(\frac{N_k}{N}\right)^2 \widehat{\mathrm{Var}}(\widehat{\tau}_{[k]})$$

- It does not take into account the uncertainty in estimating $e(X)$

- This often leads to an overestimation of the true variance since a surprising result in the literature is that using the estimated propensity scores decreases the asymptotic variance for estimating the average causal effect
  - Hirano et al. (2003); we will revisit this issue again

- Or use bootstrap

Example: NHEFS

- `causaldata::nhefs_complete`:

  National Health and Nutrition Examination Survey Epidemiologic Follow-up Study

Role	Variable	Description	Example Values / Notes
X	sex	Sex	0 = Female, 1 = Male
	race	Race	1 = White, 2 = Black, 3 = Other
	age	Age at baseline	e.g., 25–74
	school	Years of schooling	e.g., 0–20
	smokeintensity	Number of cigarettes smoked per day in 1971	e.g., 0–60
	smokeyrs	Years of smoking	e.g., 0–50
	exercise	Recreational exercise in 1971	0 = much, 1 = moderate, 2 = little/no
	active	Usual daily activity in 1971	0 = very active, 1 = moderately, 2 = inactive, 3 = missing
	wt71	Weight in kilograms in 1971	e.g., 45–120
A	qsmk	Quit smoking between 1st questionnaire and 1982	0 = No, 1 = Yes
Y	wt82	Weight in kilograms in 1982	e.g., 45–130
	wt82_71	Weight change (wt82 - wt71)	Can be negative or positive

- Study the effect of quit smoking on weight in 1982

Example: NHEFS

```
PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")

tau.S <- function(PS.Cut){
  K <- length(PS.Cut)-1
  Index <- list()
  for(tt in 1:K){
    Index[[tt]] <- which(PS.Cut[tt]<=PS.Est & PS.Est<PS.Cut[tt+1])
  }
  Nk  <- sapply(1:K,function(tt){ length(Index[[tt]]) })
  Nk1 <- sapply(1:K,function(tt){ sum(data$qsmk[Index[[tt]]]) })
  Nk0 <- sapply(1:K,function(tt){ sum(1-data$qsmk[Index[[tt]]]) })

  tau.k <- var.k <- rep(0,K)
  for(tt in 1:K){
    A.tt <- data$qsmk[Index[[tt]]]; Y.tt <- data$wt82[Index[[tt]]]
    LM <- lm(Y.tt~A.tt)
    tau.k[tt] <- coef(summary(LM))[2,1]
    var.k[tt] <- coef(summary(LM))[2,2]^2
  }

  Result <- list(ATE = sum(Nk/sum(Nk)*tau.k), VAR = sum((Nk/sum(Nk))^2*var.k))
  Result$CI <- c(Result$ATE-1.96*sqrt(Result$VAR), Result$ATE+1.96*sqrt(Result$VAR))
  return( Result )
}
```

Example: NHEFS

```
PS.Cut.5.fold <- c(0,quantile(PS.Est, (1:4)/5),1)
PS.Cut.10.fold <- c(0,quantile(PS.Est, (1:9)/10),1)
tau.S(PS.Cut.5.fold)
tau.S(PS.Cut.10.fold)
```

- 5 fold: $\widehat{\tau}_S = 3.043$, SE $= 0.938$, 95% CI $= (1.145, 4.941)$
- 10 fold: $\widehat{\tau}_S = 3.180$, SE $= 1.004$, 95% CI $= (1.216, 5.144)$

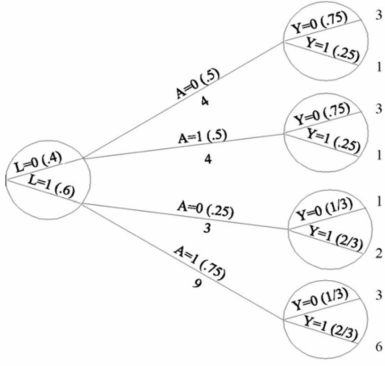
Inverse Probability Weighting

- We now focus on

$$\tau_{\mathsf{ATE}} = \mathrm{E}\left\{ \frac{AY}{\Pr(A=1|X)} - \frac{(1-A)Y}{\Pr(A=0|X)} \right\}$$

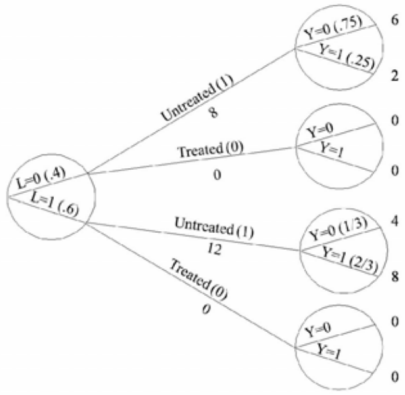
- Having proved this formula, we should now consider its interpretation

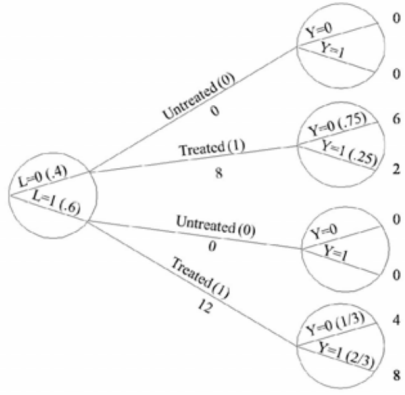
Inverse Probability Weighting



- 4 out of 8 individuals with $X = 0$ were untreated, and 1 of them died
- How many deaths would have occurred had the 8 individuals with $X = 0$ remained untreated?

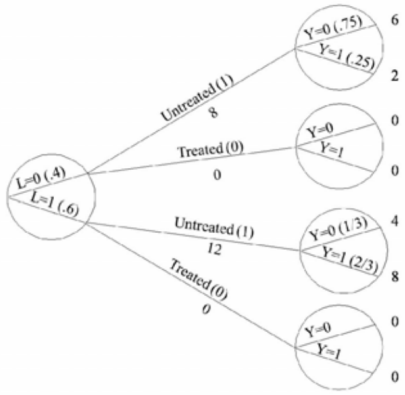
Inverse Probability Weighting

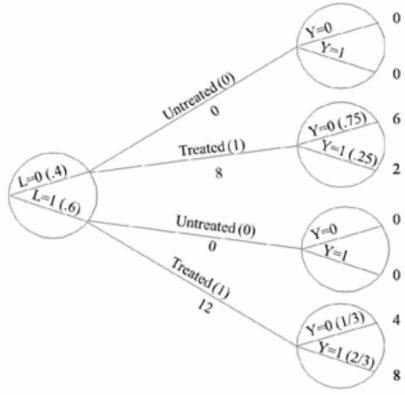




- These calculations rely on the condition that treated individuals with $X = 0$, had they remained untreated, would have had the same probability of death as those who actually remained untreated

Inverse Probability Weighting

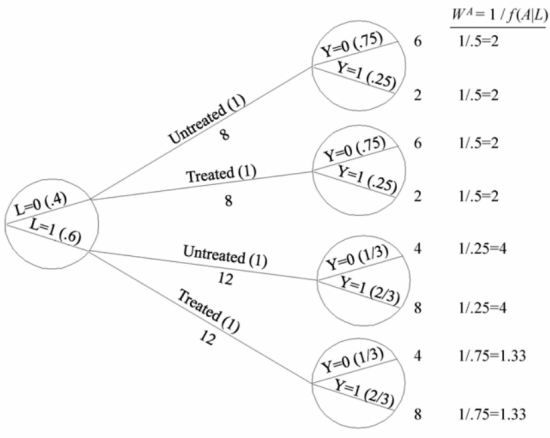




- These calculations rely on the condition that treated individuals with $X = 0$, had they remained untreated, would have had the same probability of death as those who actually remained untreated $\Rightarrow$ ignorability!

Inverse Probability Weighting

- **Idea:** create a pseudo-population
- The treatment is unconfounded in the pseudo-population



$$\Leftarrow \quad \mathrm{E}\{Y(a)\} = \mathrm{E}\left\{ \frac{\mathbb{1}(A=a)Y}{\Pr(A=a|X)} \right\}$$

IPW Estimator

- $\tau_{\text{ATE}} = \mathrm{E}\left\{ \dfrac{AY}{\Pr(A=1|X)} - \dfrac{(1-A)Y}{\Pr(A=0|X)} \right\}$

- An IPW estimator can be obtaind:

1. Estimate $e(X) = \Pr(A=1|X)$ based on statistical methods

2. Based on the plug-in principle:

$$\widehat{\tau}_{\text{IPW}} = \frac{1}{N} \sum_{i=1}^{N} \left\{ \frac{A_i Y_i}{\widehat{e}(X_i)} - \frac{(1-A_i)Y_i}{1-\widehat{e}(X_i)} \right\}$$

- The IPW estimator is also called the Horvitz–Thompson (HT) estimator (Horvitz and Thompson, 1952)

- Inference can be conducted in a manner analogous to $\widehat{\tau}_{\text{OR}}$

IPW Estimator

- $\widehat{\tau}_{\mathsf{IPW}}$ is not invariant to the location transformation of the outcome

- If we change Y_i to $Y_i + c$, $\widehat{\tau}_{\mathsf{IPW}}$ becomes

$$\widehat{\tau}_{\mathsf{IPW}} + c\left\{\frac{1}{N}\sum_{i=1}^{N}\frac{A_i}{\widehat{e}(X_i)} - \frac{1}{N}\sum_{i=1}^{N}\frac{(1-A_i)}{1-\widehat{e}(X_i)}\right\}$$

- Note that $E\left\{\dfrac{A}{e(X)} - \dfrac{1-A}{1-e(X)}\right\} = 0 \quad \not\Rightarrow \quad \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\left\{\dfrac{A_i}{\widehat{e}(X_i)} - \dfrac{(1-A_i)}{1-\widehat{e}(X_i)}\right\} \neq 0$

  - If e is estimated from a logistic regression, we have

$$\text{Score equation of the logistic regression} \quad \Rightarrow \quad \frac{1}{N}\sum_{i=1}^{N}\begin{bmatrix}1\\ X_{i1}\\ \vdots\\ X_{ip}\end{bmatrix}\widehat{e}(X_i)\{1-\widehat{e}(X_i)\}\left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{(1-A_i)}{1-\widehat{e}(X_i)}\right\} = 0$$

$$\Rightarrow \quad \frac{1}{N}\sum_{i=1}^{N}\widehat{e}(X_i)\{1-e(X_i)\}\left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{(1-A_i)}{1-\widehat{e}(X_i)}\right\} = 0$$

Hajek Estimator

- A simple fix is the Hajek estimator (Hájek, 1971): normalize the weight

$$\widehat{\tau}_{\text{Hajek}} = \frac{\sum_{i=1}^{N} \frac{A_i Y_i}{\widehat{e}(X_i)}}{\sum_{i=1}^{N} \frac{A_i}{\widehat{e}(X_i)}} - \frac{\sum_{i=1}^{N} \frac{(1-A_i) Y_i}{1-\widehat{e}(X_i)}}{\sum_{i=1}^{N} \frac{1-A_i}{1-\widehat{e}(X_i)}}$$

  - Hajek estimator is invariant to the location transformation
  - Hajek estimator is much more stable than $\widehat{\tau}_{\text{IPW}}$ in finite samples

Example: NHEFS

```
library(causaldata)
data <- nhefs_complete
PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")

X <- model.matrix(PS); A <- data$qsmk; Y <- data$wt82
IPW <- mean( (A/(PS.Est) - (1-A)/(1-PS.Est))*Y )

IPW.Boot <- rep(0,1000)
for(bb in 1:1000){
  Index <- sample(nrow(data),nrow(data),replace=TRUE)
  PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data[Index,], family="binomial")
  PS.Est <- predict(PS,type="response")
  A <- data[Index,]$qsmk; Y <- data[Index,]$wt82
  IPW.Boot[bb] <- mean( (A/(PS.Est) - (1-A)/(1-PS.Est))*Y )
}
c(IPW, sd(IPW.Boot), quantile(IPW.Boot,c(0.025,0.975)))
```

- $\widehat{\tau}_{\mathsf{IPW}} = 2.778$, SE $= 0.829$, 95% CI $= (1.259, 4.419)$

Example: NHEFS

```
library(causaldata)
data <- nhefs_complete
PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")

X <- model.matrix(PS); A <- data$qsmk; Y <- data$wt82
IPW <- mean( (A/(PS.Est) - (1-A)/(1-PS.Est))*Y )

IPW.Boot <- rep(0,1000)
for(bb in 1:1000){
  Index <- sample(nrow(data),nrow(data),replace=TRUE)
  PS.Boot <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,
                 data=data[Index,], family="binomial")
  PS.Boot.Est <- predict(PS.Boot,type="response")
  A.Boot <- data[Index,]$qsmk; Y.Boot <- data[Index,]$wt82
  IPW.Boot[bb] <- mean( (A.Boot/(PS.Boot.Est) - (1-A.Boot)/(1-PS.Boot.Est))*Y.Boot )
}
round(c(IPW, sd(IPW.Boot), quantile(IPW.Boot,c(0.025,0.975))),3)
```

- $\widehat{\tau}_{\text{IPW}} = 2.778$, SE $= 0.829$, 95% CI $= (1.259, 4.419)$

Example: NHEFS

```
MomentFts <- function(theta){ # theta[1] = ATE; theta[-1] = logistic reg. parameters
  ATE <- theta[1]; eta <- theta[-1]; PS.Est.eta <- exp(X%*%eta)/(1+exp(X%*%eta))
  residual.ATE <- ((A)/(PS.Est.eta) - (1-A)/(1-PS.Est.eta))*Y - ATE
  residual.PS <- X * matrix(PS.Est.eta*(1-PS.Est.eta)*((A)/(PS.Est.eta) - (1-A)/(1-PS.Est.eta)), nrow(X),ncol(X))
  cbind(residual.ATE,residual.PS)
}
theta.Est <- c(IPW,coef(PS))
round(apply(MomentFts(theta.Est),2,mean), 3)
```

$$\frac{1}{N}\sum_{i=1}^{N}\Psi(O_i;\widehat{\theta})=0\ , \qquad \Psi(O;\theta)=\Psi(O;(\tau,\eta))=\begin{bmatrix} \dfrac{AY}{e(X;\eta)}-\dfrac{(1-A)Y}{1-e(X;\eta)}-\tau \\ \begin{pmatrix}1\\X\end{pmatrix} e(X;\eta)\{1-e(X;\eta)\}\left\{\dfrac{A}{e(X;\eta)}-\dfrac{1-A}{1-e(X;\eta)}\right\} \end{bmatrix}$$

Example: NHEFS

```
MomentGrad <- function(theta){
  Psi <- MomentFts(theta)
  G <- matrix(0,ncol(Psi),length(theta))  # number of moments * number of parameters
  eps <- 10^(-6)
  for(tt in 1:length(theta)){
    theta.plus <- theta.minus <- theta; theta.plus[tt] <- theta[tt]+eps; theta.minus[tt] <- theta[tt]-eps
     G[,tt] <- apply( (MomentFts(theta.plus)-MomentFts(theta.minus))/(2*eps), 2, mean)
  }
  return(G)
}

MomentProduct <- function(theta){
  Psi <- MomentFts(theta);   t(Psi)%*%(Psi)/nrow(X)
}

Sigma1 <- MomentGrad(theta.Est); Sigma2 <- MomentProduct(theta.Est); Asymp.Var <- solve(Sigma1)%*%Sigma2%*%t(solve(Sigma1))/nrow(X)
SE <- Asymp.Var[1,1]^(0.5)
c(IPW, SE, IPW-1.96*SE, IPW+1.96*SE)
```

$$\widehat{\mathrm{Var}}(\widehat{\theta}) = \frac{\widehat{\Sigma}_1^{-1}\widehat{\Sigma}_2\widehat{\Sigma}_1^{-\intercal}}{N} \quad , \quad \widehat{\Sigma}_1 = \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \Psi(O_i;\widehat{\theta}) \in \mathbb{R}^{\dim(\Psi)\times\dim(\theta)} \quad , \quad \widehat{\Sigma}_2 = \frac{1}{N}\sum_{i=1}^{N} \Psi^{\otimes 2}(O_i;\widehat{\theta}) \in \mathbb{R}^{\dim(\Psi)\times\dim(\Psi)}$$

- $\widehat{\tau}_{\mathsf{IPW}} = 2.778$, SE $= 0.749$, 95% CI $= (1.310, 4.246)$

Example: NHEFS

```
PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")

X <- model.matrix(PS); A <- data$qsmk; Y <- data$wt82
W1 <- mean(A/(PS.Est))
W0 <- mean((1-A)/(1-PS.Est))
Hajek <- mean( (A/(PS.Est)/W1 - (1-A)/(1-PS.Est)/W0)*Y )

Hajek.Boot <- rep(0,1000)
for(bb in 1:1000){
  Index <- sample(nrow(data),nrow(data),replace=TRUE)
  PS.Boot <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,
                 data=data[Index,], family="binomial")
  PS.Boot.Est <- predict(PS.Boot,type="response")
  A.Boot <- data[Index,]$qsmk; Y.Boot <- data[Index,]$wt82
  W1.Boot <- mean(A.Boot/(PS.Boot.Est));  W0.Boot <- mean((1-A.Boot)/(1-PS.Boot.Est))
  Hajek.Boot[bb] <- mean( (A.Boot/(PS.Boot.Est)/W1.Boot - (1-A.Boot)/(1-PS.Boot.Est)/W0.Boot)*Y.Boot )
}
round(c(Hajek, sd(Hajek.Boot), quantile(Hajek.Boot,c(0.025,0.975))),3)
```

- $\widehat{\tau}_{\text{Hajek}} = 3.116$, SE $= 0.515$, 95% CI $= (2.148, 4.149)$

IPW Estimators Using the Estimated versus the True Propensity Score

- Hirano et al. (2003): Surprisingly,

$$\widehat{\tau}_{\text{IPW}}(\widehat{e}) = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{(1-A_i)}{1-\widehat{e}(X_i)}\right\}Y_i$$

  is more efficient than

$$\widehat{\tau}_{\text{IPW}}(e) = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{A_i}{e(X_i)} - \frac{(1-A_i)}{1-e(X_i)}\right\}Y_i$$

- Even if we know the true propensity score, we'd better not use that information!

- Intuition (Rosenbaum, 1987): estimated PS corrects for chance imbalance in the sample, but true PS does not
  - the same reason that covariate adjustment in RCT outperforms the unadjusted difference-in-means estimator

Subclassification vs weighting

- Subclassification/Stratification
  - Pros: Robust to misspecification of the PS model
  - Cons: Residual bias $\Rightarrow$ inconsistency
- Weighting
  - Pros: Correct PS model $\Rightarrow$ consistent and possibly efficient estimation
  - Cons: Highly sensitive to misspecification of the PS model
- Bottom line: a bias-variance trade-off

An Illustration from Wang et al. (2016)

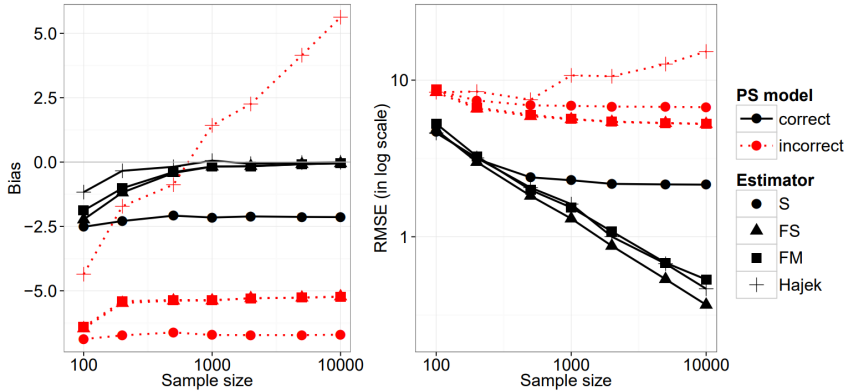


Figure 1: Bias and root mean squared error (RMSE) of the classical subclassification estimator (S), the full subclassification estimator (FS), the full matching estimator (FM) and the Hájek estimator (Hajek) under Kang and Schafer (2007)'s setting.

Subclassification $\approx$ Weighting!

- Note that the subclassification estimator is

$$\widehat{\tau}_S = \sum_{k=1}^{K} \frac{N_k}{N} \left\{ \frac{1}{N_{1k}} \sum_{i=1}^{N} A_i Y_i \mathbb{1}(\widehat{e}_i \in \widehat{C}_k) - \frac{1}{N_{0k}} \sum_{i=1}^{N} (1 - A_i) Y_i \mathbb{1}(\widehat{e}_i \in \widehat{C}_k) \right\}$$

- The subclassification estimator $\widehat{\tau}_S$ is a coarsened IPW estimator

$$\widehat{\tau}_S = \frac{1}{N} \sum_{i=1}^{N} \left\{ \frac{Y_i \mathbb{1}(A_i = 1)}{\widehat{p}_{\widehat{k}_i}} - \frac{Y_i \mathbb{1}(A_i = 0)}{1 - \widehat{p}_{\widehat{k}_i}} \right\} \qquad vs \qquad \widehat{\tau}_{\mathsf{IPW}} = \frac{1}{N} \sum_{i=1}^{N} \left\{ \frac{Y_i \mathbb{1}(A_i = 1)}{\widehat{e}_i} - \frac{Y_i \mathbb{1}(A_i = 0)}{1 - \widehat{e}_i} \right\}$$

  where we
  - estimate the propensity score (using a model): $\widehat{e}_i$
  - subclassify by quantiles of $\widehat{e}_i$'s: $\widehat{k}_i = k$ if $\widehat{e}_i \in \widehat{C}_k$
  - estimate the "subclass-specific propensity score" $p_k = \Pr(A_i = 1 \mid \widehat{e}_i \in \widehat{C}_k)$ with its empirical estimates N_{1k}/N_k
  - use $\widehat{p}_k$ as the propensity score estimates for all subject in subclass k

Bias-variance Trade-off

- The key difference between $\widehat{\tau}_{HT}$ and $\widehat{\tau}_S$ is coarsening by subclassification
  - introduce bias
  - reduce variance

- The key parameter controlling the bias-variance trade-off is the number of subclasses
  - $\widehat{\tau}_S$: $K = 5$ (typically)
  - $\widehat{\tau}_{\text{IPW}}$: $K = N$

- A larger number of subclasses would
  - reduce bias
  - increase variance

- Wang et al. (2016): proposed a hybrid estimator Select K based on a data-driven manner

Outline

1. Observational Studies
2. Preliminary
3. Propensity Score-Based Estimators
4. Balancing Estimators
5. Assess Overlap
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Balancing Property

- **Property 1**: The propensity score is the balancing score: $A \perp\!\!\!\perp X \,|\, e(X)$

- This implies

$$E\left\{\frac{Ah(X)}{e(X)}\right\} = E\left\{\frac{(1-A)h(X)}{1-e(X)}\right\}, \quad \forall h$$

- Why?

$$E\left\{\frac{Ah(X)}{e(X)}\right\} = \mathrm{E}\left[E\left\{\frac{Ah(X)}{e(X)}\,\middle|\, e(X)\right\}\right] = \mathrm{E}\left[\frac{E\{Ah(X)\,|\,e(X)\}}{e(X)}\right] = \mathrm{E}\left[\frac{E\{A\,|\,e(X)\}E\{h(X)\,|\,e(X)\}}{e(X)}\right]$$
$$\overset{(\#)}{=} \mathrm{E}\big[E\{h(X)\,|\,e(X)\}\big]$$
$$= \cdots = E\left\{\frac{(1-A)h(X)}{1-e(X)}\right\}$$

and $(\#)$ holds from

$$\Pr\{A=1 \mid e(X)\} = \mathrm{E}\{A \mid e(X)\} = \mathrm{E}\left[\mathrm{E}\left\{A \mid e(X), X\right\} \mid e(X)\right] = \mathrm{E}\left\{\mathrm{E}\left(A \mid X\right) \mid e(X)\right\} = \mathrm{E}\left\{e(X) \mid e(X)\right\} = e(X)$$

Covariate balance check

- Before accessing outcome data, we can verify whether the propensity score model is specified well enough to ensure covariate balance

- For the stratification estimator, we check:

$$A \perp\!\!\!\perp X \mid \hat{e}'(X) = e_k, \quad k = 1, \ldots, K$$

- This ensures covariate distributions are similar between treatment and control groups within each stratum

- For the IPW estimator, we view $h(X)$ as a pseudo outcome and estimate the ATE of A on $h(X)$ conditioning on $\hat{e}(X)$:

$$\texttt{lm}(h \sim A + e)$$

- Since the true effect on $h(X)$ is 0, the estimate should not significantly differ from 0

- A common choice for $h(X)$ is simply X

Example: NHEFS

```
PS <- glm(qsmk~sex+race+age+school+
            smokeintensity+smokeyrs+exercise+active+wt71,
          data=data, family="binomial")
PS.Est <- predict(PS,type="response")

X <- model.matrix(PS)

LM <- lm(X~data$qsmk+PS.Est)
Coef <- coef(summary(LM))
Summary <- matrix(0,length(Coef),5)
Summary[,1] <- names(Coef)
for(tt in 1:length(Coef)){
  Summary[tt,2:5] <- round(Coef[[tt]][2,],3)
}

colnames(Summary) <- c("Variable","Coef","SE","t-stat","p-value")
Summary
```

Variable	Coef	SE	t.stat	p.value
(Intercept)	0.000	0.000	0.276	0.782
sex1	-0.001	0.028	-0.024	0.981
race1	0.000	0.019	-0.026	0.979
age	0.029	0.608	0.048	0.961
school	0.001	0.182	0.007	0.994
smokeintensity	-0.023	0.640	-0.035	0.972
smokeyrs	0.017	0.682	0.025	0.980
exercise1	0.000	0.029	0.006	0.995
exercise2	0.000	0.029	0.008	0.993
active1	0.000	0.030	0.004	0.997
active2	0.000	0.017	0.011	0.991
wt71	0.018	0.883	0.020	0.984

Covariate Balancing Propensity Score

- Imai and Ratkovic (2014): From the balancing property of the PS, we can estimate it:

$$0 = E\left[\left\{\frac{A}{e(X)} - \frac{(1-A)}{1-e(X)}\right\} h(X)\right] \qquad \forall h$$

$$\Rightarrow \quad \text{Find} \quad \widehat{e}_B \quad \text{s.t.} \quad 0 = \frac{1}{N}\sum_{i=1}^{N}\left\{\frac{A_i}{\widehat{e}_B(X_i)} - \frac{(1-A_i)}{1-\widehat{e}_B(X_i)}\right\} h(X_i) \qquad \text{for a user-specified } h$$

- We refer to $\widehat{e}_B$ as the covariate balancing propensity score (CBPS)

- Logistic regression v. CBPS

$$\frac{1}{N}\sum_{i=1}^{N} X_i \widehat{e}(X_i)\{1-\widehat{e}(X_i)\}\left\{\frac{A_i}{\widehat{e}(X_i)} - \frac{(1-A_i)}{1-\widehat{e}(X_i)}\right\} = 0 \qquad \frac{1}{N}\sum_{i=1}^{N} h(X_i)\left\{\frac{A_i}{\widehat{e}_B(X_i)} - \frac{(1-A_i)}{1-\widehat{e}_B(X_i)}\right\} = 0$$

  - Note that $h(X)$ does not vary w.r.t. the propensity score estimate

- Now use $\widehat{e}_B$ in the IPW estimator

Example: NHEFS

```
library(WeightIt)

CBPS <- weightit(qsmk~sex+race+age+school+
                   smokeintensity+smokeyrs+exercise+active+wt71,
                 data=data,
                 method="CBPS",
                 estimand="ATE")
X.contrast <- (data$qsmk/CBPS$ps - (1-data$qsmk)/(1-CBPS$ps))*model.matrix(CBPS$formula,data)
round( apply(X.contrast,2,function(v){t.test(v)$statistic}), 4)
```

```
   (Intercept)           sex1          race1            age         school smokeintensity       smokeyrs      exercise1
             0              0              0              0              0              0              0              0
     exercise2        active1        active2           wt71
             0              0              0              0
```

Example: NHEFS

```
ATE.CBPS <- mean( (data$qsmk/CBPS$ps - (1-data$qsmk)/(1-CBPS$ps))*data$wt82 )
ATE.Boot.CBPS <- rep(0,100)
for(bb in 1:100){
  Index <- sample(nrow(data),nrow(data),replace=TRUE)
  CBPS <- weightit(qsmk~sex+race+age+school+
                     smokeintensity+smokeyrs+exercise+active+wt71,
                   data=data[Index,],
                   method="CBPS",
                   estimand="ATE")
  ATE.Boot.CBPS[bb] <- mean( (data[Index,]$qsmk/CBPS$ps - (1-data[Index,]$qsmk)/(1-CBPS$ps))*data[Index,]$wt82 )
}

round(c(ATE.CBPS,sd(ATE.Boot.CBPS),quantile(ATE.Boot.CBPS,c(0.025,0.975))),3)
```

- $\widehat{\tau}_{\text{CBPS}} = 3.290$, SE $= 0.510$, 95% CI $= (2.469, 4.298)$

Distributional Balancing

- The CBPS depends on the specified h.
  For example, if $h(X) = X$, the CBPS guarantees balance in terms of the sample mean of X

- However, it may fail to achieve balance with respect to higher-order terms or any nonlinear function of X

- Distributional balancing seeks weights that make the treated and control group distributions equal:

$$\text{Find} \quad (\widehat{w}_1, \ldots, \widehat{w}_N) \quad \text{s.t.} \quad P_{1,\widehat{w}} \simeq P_{0,\widehat{w}}$$

  where $P_{a,w}$ is the weighted distribution of the covariates in $A = a$ group with the empirical CDF:

$$F_{a,w}(x) = \frac{1}{N_a} \sum_{i=1}^{N} \mathbb{1}(A_i = a) w_i \mathbb{1}(X_i \leqslant x)$$

- More formally:

$$\text{Find} \quad (\widehat{w}_1, \ldots, \widehat{w}_N) = \underset{w \in \mathbb{R}^N}{\arg\min}\, \mathcal{D}\big(P_{1,\widehat{w}}, P_{0,\widehat{w}}\big)$$

  where $\mathcal{D}(P, Q)$ is the distance between two distributions P and Q (Wasserstein distance, Maximum mean discrepancy, etc.)

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Overlap Assumption

- (Weak) Overlap Assumption:

$$\Pr(A = a|X) \in (0, 1) \ , \quad \forall(a, X)$$

- Empirical counterpart:

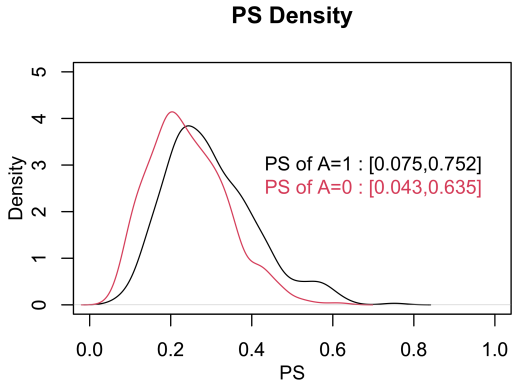
$$0 < \widehat{\Pr}(A_i = a \mid X_i) < 1 \quad \forall(a, i)$$

Example: NHEFS

```
PS <- glm(qsmk~sex+race+age+school+
            smokeintensity+smokeyrs+exercise+active+wt71,
          data=data, family="binomial")
PS.Est <- predict(PS,type="response")

PS.1 <- density(PS.Est[data$qsmk==1])
PS.0 <- density(PS.Est[data$qsmk==0])

plot(PS.1,ylim=c(0,5),xlim=c(0,1),
     xlab="",ylab="",main="")
par(new=TRUE)
plot(PS.0,col=2,ylim=c(0,5),xlim=c(0,1),
     xlab="",ylab="",main="")
title(xlab="PS",ylab="Density",line=2)
title(main="PS Density")
text(0.7,3,sprintf("PS of A=1 : [%0.3f,%0.3f]",
                   min(PS.Est[data$qsmk==1]),
                   max(PS.Est[data$qsmk==1])),col=1)
text(0.7,2.5,sprintf("PS of A=0 : [%0.3f,%0.3f]",
                     min(PS.Est[data$qsmk==0]),
                     max(PS.Est[data$qsmk==0])),col=2)
```



Truncation and Trimming

- The estimated propensity scores can be close to 0 or 1. When this happens, the weighting estimators blow up to infinity, which results in extremely unstable behavior in finite samples

- We can either winsorize the estimated propensity score:

$$\widehat{e}^{W}(X_i) = \begin{cases} \xi & \text{if } \widehat{e}(X_i) < \xi \\ \widehat{e}(X_i) & \text{if } \widehat{e}(X_i) \in [\xi, 1-\xi] \\ 1-\xi & \text{if } \widehat{e}(X_i) > 1-\xi \end{cases}$$

- or trim the observations by dropping units with $\widehat{e}(X_i)$ outside the interval $[\xi_L, \xi_U]$

- Overall, although truncation/trimming often stabilizes the IPW estimators, it also brings additional arbitrariness into the procedure

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Assignment

- Analyze the dataset `causalweight::JC` (Job Corps)
  While the dataset contains additional variables, we will focus only on the following:
  - $X =$ (`female, age, white, black, hispanic`)
  - $A =$ `assignment`
  - $Y =$ `earny4`

  We aim to study the effect of Job Corps program assignment on weekly earnings in the fourth year after assignment

- Using the dataset, obtain the ATE estimates based on the following:
  - the startification estimator;
  - the IPW estimator;
  - the Hajek estimator;
  - the CBPS-IPW estimator;

- For the IPW and CBPS-IPW estimators, assess overlap and covariate balance

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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